PRISM: an auditable, phase-aware pipeline for post-treatment glioma surveillance — rigorous evaluation of imaging, clinical and treatment-course information

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Abstract

Background. PRISM (Post-treatment Response Intelligence and Surveillance Monitor), our multimodal pipeline, predicts documented progression after a glioma surveillance MRI and explains it in a clinician report. Its first version scored AUC 0.80 on one 30-patient split of MU-Glioma-Post.

Purpose. To present PRISM, evaluate every branch and explain the optimistic initial result.

Methods. PRISM reads clinical, molecular and treatment-course data as of the scan day, adds volumes, radiomics and a frozen SwinUNETR encoder, and writes an audit record and a grounded report per scan. Development: MU-Glioma-Post (308 index scans, 155 patients; grade 4: 249 scans, 126 patients); locked external evaluation: LUMIERE (132 scans, 61 patients). Outcome: documented progression within 120 days (death censored; IPCW AUC). We ran 500 splits of the initial split's design and ablated each branch under 5×5 patient-grouped nested cross-validation; key analyses had pre-specified rules.

Results. On the initial split, PRISM's initial radiomics model scored 0.802 and a model without imaging 0.871; with treatment dates leaking from the future, a boosting model reached 0.901. Over 500 splits its pipeline averaged 0.720 ± 0.055, and 7% of splits reached 0.802. Under

repeated cross-validation the clinical + treatment-phase core reached 0.698 (95% CI 0.615–0.774) in grade 4. Radiomics changed this by −0.032 (−0.084 to +0.017) and deep features by −0.004 (−0.069 to +0.063). In post-radiotherapy scans, imaging added +0.061 (−0.038 to +0.156; permutation p = 0.21). Locked external AUCs were 0.608–0.620. The audit layer met its coverage target (0.841, 0.790–0.885), did not improve discrimination beyond case mix, and exposed a missingness shortcut. Local-language-model reports were faithful to the record in 236/249 scans (0.948; target 0.95 missed by one report, all failures one missing citation), with no omitted flag or forbidden content.

Conclusion. PRISM's signal comes from phase of care, not MRI texture. It is deployed as the clinical + treatment-phase core with auditable records; imaging branches are optional.

Key points

- **Question.** Which parts of a multimodal surveillance pipeline (PRISM) predict documented progression within 120 days of a post-treatment glioma MRI, and how far did our own initial single-split evaluation mislead?
- **Findings.** On our favourable 29-patient initial evaluation split, models scored 0.80–0.90; the same pipelines averaged about 0.70 under repeated cross-validation, and PRISM's initial radiomics pipeline averaged 0.720 over 500 splits of the same design. Radiomics, frozen deep features and re-segmentation added no measurable discrimination to PRISM's clinical + treatment-phase core (AUC 0.698 in grade 4), internally or on an external cohort.
- **Meaning.** PRISM is deployed as its clinical + treatment-phase core. Its audit records and grounded reports state what was known at the scan date, the uncertainty and the weak points of each prediction, but abstention does not triage scans better than case mix does. Single-split AUCs on cohorts of this size measure the split.

Clinical relevance statement

At a surveillance MRI after glioma treatment, PRISM's core — age, molecular markers and where the scan sits in the treatment course — predicts documented progression within 120 days about as well as any MRI-based branch tested here (AUC about 0.70), and each prediction comes with an audit record of its inputs, interval and referral flags. Conventional MRI radiomics should not be expected to add to it, and a single test-split AUC near 0.80, including our own initial one, should not be taken at face value.

1 Introduction

Most patients with high-grade glioma progress after surgery and chemoradiotherapy. They are followed with MRI every few months. Each surveillance scan raises the same question: how likely is it that progression will be documented before the next decision point? The question is hard for two reasons. First, enhancement can increase after radiotherapy without tumour growth. This pseudoprogression is most common in the first 12 weeks after chemoradiotherapy and is more frequent with MGMT promoter methylation (Brandes et al., 2008; Taal et al., 2008). The RANO criteria therefore treat early post-radiotherapy changes with caution (Wen et al., 2010), and RANO 2.0 asks for confirmation of progression within the first 12 weeks (Wen et al., 2023). Second, "progression" in routine data is a clinical documentation event. It depends on imaging, on symptoms and on when the next scan is scheduled. A model that predicts it is a prognostic model for a pragmatic endpoint, and its intended user is the neuro-oncology team deciding whether to shorten the interval to the next scan or to act on an equivocal finding. For that use, a well-calibrated probability with an honest statement of uncertainty, and a way to check what the model knew, matter as much as a high AUC.

Radiomics and deep learning have been proposed to answer this question from the images. Most work addresses an adjacent task, the separation of pseudoprogression from true progression at a lesion that is already suspicious. Here pooled AUCs are about 0.77 for machine-learning models (Booth et al., 2022), and diffusion and perfusion measures reach AUC 0.90–0.93 in meta-analysis (Taylor et al., 2022). Forward prediction from every surveillance scan has received less attention. On the public MU-Glioma-Post dataset (Mahmoud et al., 2025), a calibrated radiomics model reported AUC 0.80 on one held-out patient set (Christodoulou et al., 2026), and a nested-CV model of volume increase at the next scan reported 0.728 (Rai, 2026) [CHECK: Rai 2026 is "repo, unverified (this pass)" in COMPARABLE_RESULTS.md]. Neither returns a structured, checkable account of a prediction to the treating team.

Three problems are well documented in this literature. First, single train–test splits are unstable in small radiomics cohorts. Over 1,000 random splits of a brain-tumour radiomics dataset, one split gave 0.882 in training and 0.667 in testing, and another 0.709 and 0.911 (An et al., 2021). Cross-validation error bars are about ±10% at n = 100 (Varoquaux, 2018). Second, performance drops on external data: 81% of externally validated deep-learning algorithms lost performance, and 24% lost at least 0.10 (Yu et al., 2022), and only 29.4% of neuro-oncology radiomics studies were externally validated (Park et al., 2020). Third, clinical-only comparators are rarely reported. Where they are, they are competitive. In pseudoprogression classification, clinical variables with MGMT reached 0.69 and adding radiomics made the model significantly worse (Leone et al., 2023). A clinical-only model reached 0.727 externally against 0.717 for imaging alone (Gomaa et al., 2025). In survival prediction from pre-treatment MRI, clinical variables reached 0.74 against 0.75 for the combined radiomics model (Abler et al., 2025). These

are different tasks, often with diffusion or perfusion inputs, so their AUCs are context rather than like-for-like comparators. One study found the reverse: a clinical-only random forest scored 0.52 on a 19-patient test set (Jang et al., 2018).

A fourth problem has received little attention. In the two best-documented clinical comparators, time since the end of radiotherapy was an input, and in one it was the most influential feature (Gomaa et al., 2025; Leone et al., 2023). A surveillance cohort mixes scans taken before radiotherapy, during it, in the 12-week pseudoprogression window and in later follow-up. Event rates differ sharply between these phases. A model that only recognises the phase can rank scans well when the test set contains many easy phases. We found no study that treats phase of care as a confounder of imaging-model AUC for forward progression prediction. A related issue is leakage of information that is not yet available at prediction time (Kapoor & Narayanan, 2023). In routine clinical tables, treatment fields are recorded once per patient. If they are read as of the end of follow-up, a scan taken before radiotherapy "knows" the radiotherapy that follows it, and the model learns the phase of care from the future.

We built PRISM for this setting with three design goals: (1) *prediction* — a calibrated probability of documented progression within 120 days from everything available on the scan day: clinical and molecular data, the treatment course, lesion volumes, radiomics and deep image features; (2) *interpretability* — each prediction decomposed into the contributions of its inputs and placed in the phase of care; and (3) an *auditable clinician report* in which every statement can be traced to a recorded value. The first version (PRISM v1) combined a calibrated radiomics model, a transformer image encoder and a late-fusion network with an agentic report generator, and scored AUC 0.80 on one 30-patient split of MU-Glioma-Post, which we call our initial evaluation split. Before building on that number, we tested it, and the three problems above — single-split variance, phase of care and leakage — all turned out to apply to it.

This paper presents PRISM and its rigorous evaluation. Our contributions are: (i) the PRISM pipeline, rebuilt so that every input is read as of the scan day, every branch can be ablated, and every prediction produces an audit record and a report grounded in that record; (ii) a transparent self-correction of our initial single-split evaluation, with a split-variance and composition analysis that shows why it was optimistic; (iii) a branch-by-branch ablation of clinical, treatment-phase and imaging information under nested, patient-grouped, censoring-aware evaluation with internal, external and third-cohort testing, which identifies phase of care as the driver of discrimination; and (iv) an evaluation of PRISM's audit layer and report generator against pre-specified criteria, including what they cannot do.

2 Methods

2.1 PRISM: architecture and data flow

PRISM takes one post-treatment MRI (T1, T1c, T2 and FLAIR with a four-compartment mask) and the patient's clinical tables, and returns a risk of documented progression within 120 days, an audit record and a clinician report (Figure 5A). Every input is read as of the scan day. Five predictor branches feed one model (§2.5): (1) clinical and molecular data; (2) the treatment course and phase of care, from timing since diagnosis and radiotherapy (RT) and as-of-scan treatment status; (3) imaging measurements (volumes, morphology, change); (4) radiomics; and (5) a frozen SwinUNETR image encoder. Radiomics and deep features enter either concatenated or as a stacked score (late fusion). The predictor is a ridge logistic model over the enabled branches; every fitted step, including feature selection and branch integration, is fitted inside the training data (§2.8). Downstream, the **audit layer** wraps the deployed predictor and writes a per-scan audit record with provenance, phase of care, a risk interval, exact contributions, a conformal prediction set and referral flags (§2.12). The **report generator** turns each record into a clinician report: deterministic sections rendered from the record and a language-model summary in which every sentence cites the record fields it uses (§2.13).

PRISM v1 and what changed. PRISM v1 differed in four ways. Its radiomics branch, *PRISM's initial radiomics model*, was a calibrated logistic regression on T1c and FLAIR radiomics plus clinical codes (§2.6). Its image encoder was a small SwinUNETR-style transformer with random, untrained weights. A late-fusion MLP combined the encoder embedding with the radiomics score (*the initial PRISM fusion pipeline*). Its report generator combined hard-coded RANO 2.0 rules, a RANO–model concordance check and a large language model (GPT-4.1-mini). When we reproduced v1, the fusion MLP did not exceed its radiomics input alone (AUC 0.8006 against 0.8043 in v1's own scoring), the random encoder carried no outcome signal (alone, AUC about 0.50), and the report's RANO measurements were derived from the model score rather than from the masks (Supplementary S7). We therefore replaced the encoder with pretrained frozen encoders tested against a random-weight control, replaced the fusion MLP with one regularised model in which each branch is an ablatable block, added the as-of-scan rule and censoring-aware labels, and rebuilt the report generator on the audit record.

Deployed configuration. A branch enters the deployed model only if it improves cross-validated discrimination. After the evaluation below, the deployed PRISM configuration is the *core model*: branches 1–2 (clinical + timing + as-of-scan treatment; 23 predictors, 31 design columns; ridge). The core model is also the reference ("baseline") for every imaging comparison. Branches 3–5 remain optional modules for research use; they add no measurable discrimination on these data.

2.2 Study design and reporting

This retrospective study of public, de-identified data develops and internally evaluates PRISM on one cohort (MU-Glioma-Post), evaluates locked models on a second (LUMIERE) and runs a supporting analysis on a third (CFB-GBM). Reporting follows TRIPOD+AI (Collins et al., 2024) and CLEAR (Kocak et al., 2023 [CHECK]); both checklists are cross-referenced in Supplementary S17. Analyses labelled *pre-specified* had their plan and decision rule written into the analysis document and code before any model output of that analysis was seen; for the clean-label, audit and CFB-GBM analyses a dated plan snapshot was saved before the run (Supplementary S1). Everything else is secondary or exploratory.

2.3 Data sources

MU-Glioma-Post (development; Mahmoud et al., 2025) is a single-centre collection of post-operative MRI (T1, T1c, T2, FLAIR) with four-compartment masks (non-enhancing tumour core, NETC; surrounding FLAIR hyperintensity, SNFH; enhancing tumour, ET; resection cavity, RC) and a clinical workbook. The workbook gives days from diagnosis to each MRI, surgery, RT and chemotherapy start and end, first progression and death, plus age, sex, grade, IDH, MGMT and the type of first progression (imaging, clinical or both). Images carry no acquisition metadata. **LUMIERE** (external; Suter et al., 2022) is a single-centre longitudinal glioblastoma collection with an expert RANO rating per follow-up, automatic DeepBraTumIA segmentations, IDH and MGMT, but no RT or chemotherapy dates; its only treatment-time variable is a "< 3 months after RT" flag. **CFB-GBM** (supporting; TCIA doi:10.7937/v9pn-2f72 [CHECK]) contains 264 patients treated with the Stupp protocol, imaged at RT planning (t0) and twice afterwards (t1, t2), including ADC, with volumetric RANO 2.0 labels and precomputed PyRadiomics features on the gross tumour volume (GTV), but no IDH or MGMT. We used only its released tables.

2.4 Index scans and outcome

An index scan is a post-treatment MRI before first documented progression and before late treatment, with all four sequences and a mask, a known and plausible MRI day, and no recorded death before it. The flow from 596 timepoints of 203 patients to 308 index scans of 155 patients is in Supplementary S2. The grade 4 subset (249 scans, 126 patients) is the pre-specified primary population. LUMIERE index scans are rated follow-ups at least two weeks after surgery, before the first rating of progressive disease. Each patient contributes all eligible scans, and every analysis keeps a patient's scans together.

The primary outcome is documented first progression within $H = 120$ days of the index scan (90 and 180 days as sensitivity analyses). A scan is negative only if progression-free follow-up reaches

120 days; otherwise it is censored. Death without documented progression censors at the death day. Rules for conflicting dates were fixed in an audit before the final run; six index scans left the cohort, and at 120 days 17 negatives became censored and one became positive (Supplementary S3). In LUMIERE, progression is the first expert rating of progressive disease. PRISM v1 used a "naive" label that counted censored scans as negatives; we use it only where we reproduce the initial evaluation. Models are fitted on scans whose 120-day status is known. Discrimination is the inverse-probability-of-censoring-weighted (IPCW) cumulative/dynamic AUC at 120 days (Uno et al., 2007; Blanche et al., 2013). In a planned comparison of label formulations, none changed discrimination (all 36 ΔAUC CIs included 0), but the naive label under-predicted risk (observed/expected, O/E, 1.12–1.19 against 0.95–0.99; Supplementary S3).

2.5 Predictors: PRISM's input branches

All predictors are defined as of the scan day (definitions in Supplementary S4).

- **Clinical** (branch 1): age, sex, IDH (wild type, mutant, unknown), MGMT (methylated, unmethylated, unknown).
- **Timing** (branch 2): days from diagnosis, days since RT end, an early post-RT flag (0–84 days after RT end, the RANO 12-week window) and its interaction with MGMT methylation, visit index, and an unknown-RT-end flag.
- **Treatment, as of the scan** (branch 2): whether RT and chemotherapy had started, whether RT was completed, prescription, RT start day, RT duration, surgery day, biopsy before resection and previous brain tumour. A treatment fact enters only once its documented start day is on or before the scan day; RT duration and days since RT end appear only after RT has ended. Unknown values are imputed in-fold with missingness indicators. The **core model** ("baseline") is clinical + timing + treatment (23 predictors, 31 design columns).
- **Phase of care** (branch 2, compact form): pre-RT, during RT, early post-RT ($\leq$ 84 days) or later post-RT, plus log days since RT end.
- **Imaging measurements** (branch 3): compartment volumes; morphology (ET fraction, multifocality, ET-to-cavity distance, intensities relative to normal brain); and past-only change from the previous scan and the nadir.
- **Radiomics** (branch 4): PRISM v1's legacy T1c + FLAIR features (2,446 columns) and corrected features (whole-brain z-score, fixed 32-bin discretisation, per compartment, all four sequences). In every training fold each radiomics and deep feature is clipped to that fold's 1st and 99th percentiles, which removes a PyRadiomics sentinel value (10^6) found in three legacy columns of four scans (Supplementary S5).

- **Deep features** (branch 5): frozen SwinUNETR encoders with BraTS 2021 and BrainSegFounder weights (Cox et al., 2024 [CHECK]), and the same architecture with random weights as a control (the v1 encoder type); mask-weighted pooled features per region (Supplementary S6).

2.6 PRISM's initial radiomics model, our initial evaluation split and the as-of-scan correction

PRISM's initial radiomics model is an L2-penalised logistic regression on 46 legacy radiomics and clinical-code columns with Platt calibration; its features came from an L1 + permutation consensus ranking and its hyperparameters from an Optuna search (Supplementary S7). Our initial evaluation split came from PRISM v1's own split search: for a seed, 20,000 random draws of 30 patients, keeping the draw closest to 96 test scans and 53 positives; seed 62 gives the split. We reproduced the shipped model's stored test predictions to 1.1×10^{-15}. In the label-fixed cohort the split has 126 training and 29 test patients with 82 test scans (22 naive positives). On it we fitted, once and without tuning on the split, the core ridge model, a gradient-boosting version of it, the core model plus legacy radiomics, and a clinical-only ridge. CIs and paired ΔAUC against the core ridge model come from a 1,000-draw patient-cluster bootstrap over test patients.

In our earlier runs, treatment and timing features read the workbook as of the end of follow-up. A pre-RT scan therefore carried the dose, start date and "RT given" status of a course that had not started, and a "days since RT end" computed from a future date (RT status was wrong in 23 all-glioma scans and RT duration or days since RT end in 29; Supplementary S4). We fixed this with the as-of rule above and report the split before and after the fix (Table 2).

2.7 Split variance

We ran PRISM v1's split search for seeds 0–499, giving 500 distinct 30-patient test sets (82–94 scans, 10–33 positives). On each split's training patients we refitted (a) PRISM's initial radiomics pipeline with its feature ranking redone (the honest estimate; Optuna not re-run), (b) the shipped 48 features (optimistic, because they were selected on patients who are test patients in most other splits), (c) the core ridge model and (d) the core model plus legacy radiomics. We report naive-label AUC (500 splits) and IPCW AUC (475 splits; undefined on 25), the percentile rank of our initial evaluation split, and the relation of per-split AUC to test-set composition. This analysis is descriptive; verifier checks are in Supplementary S8.

2.8 Repeated cross-validation and component ablation

All repeated-CV analyses share one set of 5 × 5 repeated patient-grouped folds stratified by 120-day status and grade. Every fitted step — imputation, clipping, scaling, selection, stacking and hyperparameters — is fitted inside the training fold (Varma & Simon, 2006; Cawley & Talbot, 2010). The headline estimate is the mean over the five repeats of the per-repeat pooled out-of-fold IPCW AUC(120 d); an earlier repeat-averaged estimator, which acts as an ensemble, was replaced before the final run (Supplementary S9). 95% CIs and paired ΔAUC come from a 1,000-draw patient-cluster bootstrap in which all models are scored on the same resampled patients.

Each branch is ablated by adding it to, or removing it from, the core model. The primary learner is ridge logistic regression with the penalty chosen by inner patient-grouped log-loss; regularised gradient boosting is secondary. The radiomics branch enters through 70 configurations (feature variant × stability filter × selection × concatenation or a stacked radiomics score; Supplementary S5). The honest estimate chooses one configuration per outer training fold by inner 3-fold patient-grouped CV (nested selection). The deep branch uses the same protocol over 32 configurations, with a 16-configuration random-weight control. With 70–80 uncorrected intervals, about 1.75–2 single configurations are expected to exclude 0 by chance. **The pre-specified primary development analysis** was grade 4: core ridge model versus core + radiomics by nested selection.

2.9 Phase of care (pre-specified)

A ladder of non-imaging bases — R0 clinical; R1 + timing; R2 clinical + phase; R3 core model — received four imaging additions: I1 volumes; I2 all non-radiomics imaging measurements; I3 radiomics (nested); I4 deep features (nested). Cohorts were the full cohort, the post-RT cohort (RT documented and completed; grade 4: 167 scans, 84 patients, 83 events) and the later post-RT cohort (> 84 days after RT end). Models were trained and evaluated within each cohort. The primary contrast was grade 4, post-RT, ridge, R2 + I2 versus R2, with a 100-permutation test that moves the I2 block between patients with the same number of scans; the key secondary contrast was R2 + I3 versus R2. The plan was written into the code before the full run but not committed separately (Supplementary S11).

2.10 Label, positive controls and re-segmentation

Imaging-defined label (pre-specified). Clinical-only first progressions were censored at the progression date. Rule: the label is "the bottleneck" if, in grade 4 (ridge), imaging alone or core + imaging (volumes + change) beats the core model by ΔAUC ≥ 0.03 with a 95% CI excluding 0. **Positive controls.** We checked volumes against the dataset's own volume table,

label mapping, label–date alignment and 12 random masks, and measured concurrent detection of the progression scan against verified 120-day negatives (imaging only). **Re-segmentation (pre-specified).** All 594 complete MU scans were re-segmented with the MONAI BraTS 2021 SwinUNETR (primary) and BrainSegFounder. The new masks "help" if concurrent detection rises by ≥ 0.05 with a CI excluding 0, or if ET is larger at the progression scan than at the previous scan in more than 65% of pairs.

2.11 External evaluation and diffusion

Models trained on all MU scans were applied unchanged to LUMIERE. Without treatment dates, the transportable features are clinical + timing (log days since surgery, the $<$ 3-month flag) $\pm$ harmonised volumes; the pre-specified external model was clinical + timing + volumes trained on grade 4. We report pooled IPCW AUC, O/E and calibration slope, and on 5×5 LUMIERE folds recalibration, fine-tuning and a cohort-normalised volume sensitivity analysis (Supplementary S13). PRISM's initial radiomics model, as shipped, was applied to legacy features re-extracted on LUMIERE.

In CFB-GBM (pre-specified; one row per patient), the outcome was volumetric RANO 2.0 progression from t1 to t2. Nested feature sets were (a) clinical, (b) + GTV volume and its change from t0, (c) + conventional T1-Gd and FLAIR radiomics, (d) + ADC radiomics, fitted by ridge with in-fold clipping and selection and evaluated by 20 repeats of 5-fold CV. Rule: ADC helps if (d) $-$ (c) ≥ 0.03 with a CI excluding 0, with a 1,000-permutation test of the ADC block. Caveats were fixed in advance: no molecular markers; t1 falls inside the 12-week window; and volume at t1 is part of the label, so (b) is partly circular.

2.12 The audit layer

The audit layer wraps the core model; it adds no new predictor (plan snapshot saved before the run; Supplementary S15). It has five parts. (1) A **locked model** with a model card: code version, hashes of every input table, each input's definition, unit, source column and as-of rule, training counts, cross-validated performance, intended use and out-of-scope uses. (2) A **per-scan audit record**: each input with its source and as-of day and a check that it was known by the scan day; phase of care; risk with a 95% interval from 200 patient-cluster bootstrap refits; exact additive log-odds contributions by group, checked against the logit; a split-conformal prediction set; and referral flags with reasons — (i) early post-RT window, (ii) out of distribution (IDH or MGMT unknown, or an input outside the training 1st–99th percentile), (iii) uncertain (conformal set $\{0, 1\}$ at $\alpha = 0.2$). (3) **Selective prediction**: risk–coverage curves compared with random removal and with composition-matched removal (same retained count per phase). (4) A **shortcut and bias audit**: refits without the "unknown" molecular indicators, timing-only and treatment-only models, subgroup calibration and a transport audit on LUMIERE. (5)

Explanation stability and a reproducibility manifest. Everything was evaluated inside the shared folds with every fitted step, including the conformal calibration split (70/30 of training patients), fitted on training patients only.

Two quantities were pre-specified. **P1:** conformal coverage at $\alpha = 0.2$ on verified test scans ≥ 0.78 overall, with per-phase coverage reported as conditional coverage. **P2:** AUC on scans retained at about 70% coverage minus the AUC of composition-matched retention, met if its 95% CI lies above 0.

2.13 The report generator and its faithfulness evaluation

The report is written only from one scan's audit record, extended with a descriptive risk tier (tertiles of the fold model's predicted risk on its own training scans; not a decision threshold) and past-only compartment volumes labelled indicative, not RANO. Sections for scan context, inputs (with source and as-of day), risk (interval, tier, conformal set), contributions, referral flags with reasons and volumetric change are rendered deterministically from the record. A summary is written by a local language model (Qwen3.8-27B at a pinned revision, bf16, run offline on the study's own GPUs, greedy decoding) whose only input is a flat view of the record's fields. The prompt allows only facts from the record, requires every number to be copied from a field and every sentence to cite the field keys it uses, requires every active flag to be stated, and forbids diagnoses, RANO categories, management advice and unhedged certainty; the output is constrained to a list of sentences with their cited keys. From PRISM v1 we kept the change-from-prior and nadir table, the explanation section (now exact contributions instead of SHAP on the fusion network), tiered confidence language and the disclaimer. We dropped the fixed alert threshold (hand-set on the initial split and supported by none of our evaluations), the RANO rule engine and RANO–model concordance (computed from the model's own score, not from measurements), an unfitted pseudoprogression confounder score, and the printing of the outcome label (Supplementary S19).

A deterministic checker scores every sentence: cited keys must exist, and every number must equal a cited field's value up to the written rounding. Per report, every active flag must be mentioned and no inactive flag described as raised, the stated tier must agree with the record, and no forbidden pattern (diagnosis, RANO category, management, unhedged certainty) may appear. Prompts and checker patterns were developed on the 59 non-grade-4 scans; the plan was frozen before any evaluation report was generated. The evaluation set was all 249 grade 4 scans, with records from the out-of-fold fits of one CV repeat. **The pre-specified primary quantity** was the proportion of grounded reports with zero unsupported claims and every active flag mentioned (Wilson 95% CI; target ≥ 0.95). Secondary arms, compared on the same scans with a 2,000-draw patient-cluster bootstrap, were the original PRISM v1 prompt run with the same model on the same facts (arm B; its numbers checked against its own input) and the grounded prompt with a smaller model (Qwen3.5-9B; arm C). A stress test perturbed 30 records (risk

moved to another tier, a flag switched on or off) and required the report to follow the change.

2.14 Further analyses and software

The Supplement reports a feature-group search with a permutation control (S10), literature baselines (including a Hashim-style Cox model; Hashim et al., 2026), hyperparameter grids, class imbalance, schedule coupling, feature conflicts, SHAP, calibration and decision curves (Van Calster et al., 2019; Pencina et al., 2008) and sample size by the Riley criteria (Riley et al., 2020). Code is Python (scikit-learn, scikit-survival, PyRadiomics, MONAI, SHAP); every stage is one command with fixed seeds, and logs, folds and hashes are kept with the outputs (Supplementary S16).

3 Results

3.1 Cohorts

MU-Glioma-Post contributed 308 index scans from 155 patients (Table 1). Most were grade 4 (126 patients, 81%), IDH wild type (77%) and treated with RT (66%). At 120 days, 100 scans were followed by documented progression, 156 were not and 52 were censored (grade 4: 93 / 117 / 39). Of the grade 4 scans, 76 were taken before RT, 6 during RT, 98 in the early post-RT window and 69 later. Event rates differed by phase: in the full cohort, 8.7% of pre-RT scans were followed by progression within 120 days. LUMIERE contributed 132 index scans from 61 glioblastoma patients (47 / 76 / 9 at 120 days). IDH status was unknown in 36% of LUMIERE patients and 8% of MU patients.

3.2 PRISM's initial single-split result and why it misled

Our initial evaluation reported AUC 0.80 for PRISM v1 on one split. Re-scored with the current cohort and labels, every model scored high on that split (Figure 1A; Table 2). PRISM's initial radiomics model reached 0.802 on the naive label and an IPCW AUC of 0.817 (95% CI 0.690–0.920). The core model, with no imaging — clinical, timing and as-of-scan treatment — reached 0.871 and 0.851 (0.735–0.947). The paired difference between the two was −0.034 (−0.186 to +0.109). Adding legacy radiomics to the core model changed it by −0.006 (−0.081 to +0.090). A clinical-only ridge model, with age, sex, IDH and MGMT alone, scored 0.791 and 0.782 (0.638–0.897). All intervals were wide and overlapping.

How a 0.70 model scored 0.90: a treatment-date leak. Before the as-of-scan correction, the same clinical + timing + treatment design fitted by gradient boosting scored 0.901 on the naive label on this split (30 test patients, 84 scans; Table 2, history rows). Under repeated CV

on the same pre-correction inputs, that model averaged 0.723 (0.633–0.806) in grade 4 and 0.743 (0.666–0.816) in all gliomas. Two things lifted it. The first was leakage: pre-RT and on-RT scans carried the RT start date, dose and end date of a course that had not yet started or ended. After the correction, the boosting model scored 0.869 on the same split, and with the final labels and the 29-patient test set, 0.844 (IPCW 0.824, 0.690–0.916). Its final repeated-CV AUC is 0.687 (0.601–0.768) in grade 4 and 0.718 (0.642–0.792) in all gliomas. The ridge model moved less: 0.870 before the correction, 0.880 after it and 0.871 with the final labels. The second lift was the split itself.

Why the split is easy: composition and phase of care. Of the 60 negative test scans, 36 were taken before RT started, and none of the 36 pre-RT test scans was followed by progression (Figure 1B). In the whole cohort 9% of such scans were. So 60% of the positive–negative pairs compared a progression scan with an early post-operative, pre-RT scan. The core model ranked these pairs with AUC 0.962 and the remaining pairs with 0.735; PRISM's initial radiomics model scored 0.823 and 0.771. Among the 500 design splits, our initial evaluation split was at the 89th percentile for the share of pre-RT scans, the 5th for IDH-mutant scans and the 6th for later post-RT scans. A no-training rule, "the scan is post-RT", scored 0.740 on this test set (92nd percentile). On the 42 post-RT test scans (19 patients), PRISM's initial radiomics model scored 0.770 (0.604–0.907) and the core model 0.718 (0.481–0.880) (Figure 1C). The clinical-only model was nominally highest there, at 0.818 (0.664–0.924), with a difference from the core model of +0.100 (−0.015 to +0.293): another illustration of single-split noise.

3.3 Split variance: 500 splits of the same design

When PRISM's initial radiomics pipeline re-selected its features on each split, its AUC averaged 0.720 ± 0.055 (2.5–97.5% range 0.610–0.819) (Figure 2A). Its re-run on our initial evaluation split scored 0.801, above 92% of splits. Only 7% of splits reached the reported 0.802. Refitting the shipped 48 features gave 0.815 ± 0.044, but this is an upper bound: those features were chosen on patients who are test patients in most other splits. Per split, the honest version was 0.095 ± 0.040 lower than the optimistic one. Re-running the full hyperparameter search did not rescue the pipeline (Supplementary S8).

The core model averaged 0.801 ± 0.052 over the same splits, and our initial evaluation split ranked it at the 91st percentile. Adding legacy radiomics gave 0.799 ± 0.053, a mean difference of −0.002; radiomics helped in 51.6% of splits. Per-split AUC tracked the phase mix of the test set (Figure 2B): for the core model, the Spearman correlation with the AUC of the "post-RT" rule was +0.57. From composition alone, nine variables predicted 0.851 for our initial evaluation split for the core model (observed 0.871, mean 0.801; CV R^2 0.42) and 0.773 for the initial radiomics pipeline (observed 0.801, mean 0.720; CV R^2 0.10). About two thirds of the split's excess over an average split is thus explained by its composition, and the rest by the particular patients.

These AUCs use the naive label and PRISM v1's case table, so their level is not comparable with repeated CV; for the same models, single-split IPCW AUCs were 0.10–0.16 above the all-glioma repeated-CV values (Table 2).

3.4 Component ablation: clinical, treatment-phase and imaging branches

In grade 4, clinical features alone reached 0.585 (0.496–0.666). Adding timing raised this to 0.644 (0.560–0.721), and adding as-of-scan treatment to 0.698 (0.615–0.774), the core model (Figure 3A). In all gliomas the steps were 0.619, 0.693 and 0.720 (0.638–0.792). Boosting did not beat ridge (grade 4 0.687; all gliomas 0.718). Tuning changed the core model by at most ±0.025, and imaging-only literature baselines were weaker (for example, a volumetric RANO-style rule 0.551; Supplementary S12).

Radiomics branch (pre-specified primary analysis). With nested selection over 70 configurations, radiomics changed the grade 4 core model by −0.032 (−0.084 to +0.017; AUC 0.666) and the all-glioma core model by +0.028 (−0.017 to +0.071; AUC 0.747) (Table 3; Figure 3B). No grade 4 configuration had a CI above 0 (8 of 70 were below 0). In all gliomas 1 of 70 did — legacy PCA-10, stacked, +0.035 (+0.002 to +0.068) — fewer than the 1.75 expected by chance. Nested radiomics models were over-confident (calibration slope 0.60 in grade 4 against 0.89 for the core model).

Deep encoder branch. Frozen pretrained encoders, nested over 32 configurations, changed the core model by −0.004 (−0.069 to +0.063) in grade 4 and +0.011 (−0.032 to +0.056) in all gliomas. The pretrained features did carry image-only signal (grade 4 image-only AUC 0.652, against 0.566 with random weights), but it correlated with tumour volume (r = 0.58) and was not additive. The random-weight control, the v1 encoder type, was worse than the core model: −0.049 (−0.085 to −0.014).

Credit is not value. In models that included volumes, morphology and change features, these imaging groups took 45% (ridge) and 51% (boosting) of the mean absolute SHAP credit, yet the models did not improve on the core model (0.692 and 0.694 against 0.698) (Figure 3C). In the core model itself, treatment took 58% of the credit, clinical 25% and timing 17%; the leading features were whether RT had started and the RT start day.

3.5 Phase of care

A compact four-level phase block recovered most of the core model's gain over clinical features: grade 4 R0 0.585, R2 0.664, R3 0.698 (R2 − R0 +0.080, +0.005 to +0.152) (Figure 4A). With phase known, volumes did as well as the full treatment block (R2 + I1 0.696) but no better. Within the post-RT cohort no model discriminated well. Grade 4 bases scored 0.549–0.604, and phase itself carried nothing (R2 − R0 −0.003).

Primary contrast (pre-specified). In grade 4 post-RT scans, adding all non-radiomics imaging measurements to clinical + phase changed AUC from 0.549 (0.458–0.642) to 0.610 (0.529–0.688): ΔAUC +0.061 (−0.038 to +0.156), permutation p = 0.21 (Figure 4B). The permutation null itself averaged +0.028: adding 44 permuted columns to a weak base raises its AUC. The key secondary contrast, radiomics, gave +0.061 (−0.041 to +0.157) with calibration slope 0.44. **The hypothesis that imaging adds value within the post-RT phase was not supported.** On top of the full core model, every imaging block changed post-RT AUC by +0.005 to +0.04 with all CIs including 0. In an exploratory check, age alone scored 0.586 in grade 4 post-RT scans, above R2, so part of the +0.061 reflects a weak base. In the later post-RT cohort (69 scans) the bases fell below 0.5 (0.343–0.472), so contrasts there are not interpretable.

3.6 Label, positive controls and segmentation

Imaging-defined label (pre-specified). Clinical-only progression accounted for 12 of 93 grade 4 events (10 patients). With these censored, core + imaging reached 0.722 (0.639–0.799) against 0.696 (0.610–0.776) for the core model: ΔAUC +0.027 (−0.033 to +0.082). Imaging alone was 0.068 below the core model. **The rule was not met**; the documented-progression label was not shown to be the bottleneck.

Positive controls. No pipeline error was found. Our volumes reproduced the dataset's own table (Pearson r 0.996– 0.9998 by compartment), labels were aligned with scan dates and the 12 inspected masks were correct. Yet imaging detected the progression scan itself against verified negatives with AUC only 0.68 (0.62–0.74). ET volume was larger at the progression scan than at the previous scan in 49% of 74 pairs (p = 0.54); whole-lesion growth from the nadir was larger in 79% (p = 0.0003). Volume series were noisy: between scans $\leq$ 14 days apart the median ET change was 3.2-fold. The 120-day label largely meant "the next scan is the progression scan" (agreement 79%, κ 0.53; Supplementary S14).

Re-segmentation (pre-specified). The BraTS 2021 SwinUNETR agreed with the MU masks (median Dice ET 0.80, whole tumour 0.85) and removed part of the short-interval noise, but primary detection fell from 0.681 to 0.665 (Δ −0.016, −0.057 to +0.025), and ET was larger at progression in 45% of pairs. BrainSegFounder gave 0.677 and 41%. **The rule was not met.** In the forward 120-day task the same clinical + phase + imaging model changed by +0.030 (−0.008 to +0.069) in grade 4 with the new masks (Table 3).

3.7 External and third-cohort evaluation

On LUMIERE the pre-specified locked model (clinical + timing + volumes, trained on MU grade 4) scored 0.608 (0.469–0.739) with ridge and 0.620 (0.495–0.733) with boosting, against 0.667 and 0.675 in MU CV. Clinical + timing alone scored 0.610 and 0.632; adding volumes

did not help (ridge −0.001, −0.073 to +0.057). The ridge model under-predicted risk (O/E 1.33); boosting was calibrated in the large (O/E 0.99). Recalibration brought O/E to 0.93–0.97 without changing the ranking, and no fine-tuning arm beat the locked model. Rank-normalising volumes within each cohort raised the boosting model to 0.659 (0.529–0.767), suggesting that part of the loss is segmenter scale. Frozen deep features added +0.050 to +0.089 (one of four CIs excluded 0), but a random-weight encoder gained up to +0.045, and deep features identified the cohort with AUC about 0.8. PRISM's initial radiomics model, as shipped, scored 0.613 (0.500–0.729) on 118 LUMIERE scans. In LUMIERE the time to the next scan alone scored 0.788 (0.703–0.868), higher than any locked model (Supplementary S13).

CFB-GBM (pre-specified). In 77 patients with ADC (19 events), adding ADC features to conventional radiomics changed AUC by +0.007 (−0.019 to +0.035; permutation p = 0.151; minimal detectable difference 0.038). **The rule was not met.** Clinical features alone were at chance (0.501 in the 107-patient conventional cohort, 0.565 in the ADC cohort). GTV volume and its change added +0.185 (+0.064 to +0.293) and +0.148 (+0.031 to +0.266), but this block is partly circular with a label defined by the same volume. Conventional texture added nothing beyond volume (−0.009 and −0.022).

3.8 Calibration and clinical utility of the core model

The grade 4 core model was well calibrated in the large (weighted O/E 0.96; calibration-in-the-large −0.09, −0.42 to 0.28) and mildly over-fitted (slope 0.89, 0.50–1.43). At a 30% threshold its net benefit was 0.232 against 0.189 for treating all. Radiomics lowered net benefit at 30% by −0.020 (−0.038 to −0.004); deep features changed it by at most ±0.002 at 10–30%. The study is under-powered by the Riley criteria even for the core model: 3.0 events per parameter for 31 design columns, against 2,190 scans (970 events) required. The radiomics result held in nine sensitivity analyses (ΔAUC −0.032 to +0.028, no CI excluding 0), and the core model held within strata of the interval to the next scan (0.687, 0.595–0.768) and against re-balancing or XGBoost (Supplementary S12).

3.9 The audit layer

The audit layer's cross-validated predictions reproduced the core model's out-of-fold predictions (maximum difference 1.7×10^{-16}), and the additive explanation reproduced every logit (maximum error 1.3×10^{-15}). An example record is in Supplementary S15.

P1 (coverage) was met overall, with conditional failures. At $\alpha = 0.2$, conformal coverage on verified test scans was 0.841 (0.790–0.885) in grade 4 and 0.823 (0.776–0.865) in all gliomas. Coverage failed in two places. Almost every pre-RT scan received the set {0}, so all 5 grade 4 pre-RT progressions (10 in all gliomas) fell outside their sets. During RT, coverage was 0.567 (6

scans).

P2 (abstention beyond case mix) was not met. At about 70% coverage, AUC on the retained scans was 0.731 (0.635–0.812) against 0.698 for random removal, but composition-matched removal gave 0.725: P2 +0.006 (−0.023 to +0.036) in grade 4 and +0.003 (−0.031 to +0.036) in all gliomas (Figure 5C). Abstention kept 97.6% of pre-RT scans. The early post-RT referral rule raised AUC to 0.723 only by removing a whole phase; within-phase AUCs were 0.555 (pre-RT), 0.663 (early post-RT) and 0.597 (later post-RT), all below the pooled 0.698. Any referral flag applied to 72.6% of grade 4 scans.

The audit found four problems that the headline AUC hides. (1) A missingness shortcut: IDH was unknown in 0 of 93 progression scans but in 6 of 117 non-progression and 5 of 39 censored scans; removing the unknown-marker indicators lowered grade 4 AUC by −0.023 (−0.047 to −0.001). (2) The core model is a treatment-course model: treatment features alone scored 0.701 (0.622–0.776), Δ +0.003 (−0.050 to +0.048) against the full core model. (3) A molecular look-ahead: IDH and MGMT were used at 8 grade 4 scans taken 1–58 days before the diagnosis date. (4) The LUMIERE miscalibration came mostly from the same shortcut: IDH was unknown in 40.9% of LUMIERE scans against 4.4% in MU, and the mean contribution of the IDH-unknown indicator shifted by −0.68 logits; refitting without the unknown indicators moved LUMIERE O/E from 1.33 to 0.89. Under this shift, conformal coverage on LUMIERE fell to 0.715 (0.340 on events). Explanation signs were stable, but the exact top 5 features recurred in at most 4% of refits.

3.10 The report generator

Primary (pre-specified) not met, by one report. Of 249 grade 4 grounded reports, 236 (0.948, Wilson 95% CI 0.913–0.969) had zero unsupported claims and mentioned every active flag, against a target of ≥ 0.95. All 13 failures had the same cause: the risk sentence stated the 120-day horizon without citing the horizon field. The number was correct, and no grounded report contained a number absent from the record. All 13 were early post-RT scans (85 of 98 passed; every pre-RT, during-RT and later post-RT report passed). Otherwise the grounded reports were clean: all 249 parsed, no citation was invalid, none of 269 active flags was omitted, no inactive flag was described as raised, no report contained forbidden content, and the tier was stated in every report and never inconsistent. Generation took 0.06 s per report in batch (median single-request latency 3.6 s).

Comparison arms. The original PRISM v1 prompt with the same model and facts (arm B) passed 171 of 249 reports against its own input (0.687, 0.627–0.741; difference +0.261, +0.197 to +0.328 in favour of the grounded prompt) and none against the record. Every arm-B report contained forbidden content (management 99%, diagnosis 51%, RANO category 41%, unhedged certainty 40%), and it omitted 48 of 269 active flags (17.8%) although they were in its input. The

grounded prompt with the 9B model (arm C) passed 243 of 249 (0.976, 0.948–0.989; difference −0.028, −0.062 to +0.008) but wrote forbidden content in 7 reports (2.8%); on reading, 1 was a real violation (a hedged diagnosis) and 6 were pattern false positives.

Stress test and reading. Of 30 perturbed records, the grounded reports passed 27 (arm B 17, arm C 28). All three failures followed the changed field and failed for other reasons: the horizon citation, a new interval bound that coincided with the old risk, and an invented field key. Reading 15 grounded reports (3 chosen by a pre-specified rule, 12 at random) against their records found no factual error; wording issues included a bootstrap interval called a "confidence interval" and a risk and tier cut point that rounded to the same value. Figure 5B shows the pre-specified early post-RT example.

4 Discussion

We present PRISM, a pipeline for post-treatment glioma surveillance that reads every input as of the scan day, places each scan in its phase of care, and returns a risk with an audit record and a grounded clinician report. Evaluating every branch rigorously led to four findings. First, our initial single-split evaluation was optimistic: over 500 splits of the same design PRISM's initial radiomics pipeline averaged 0.72, and 7% of splits reached the reported value. Second, under repeated, patient-grouped and censoring-aware evaluation, PRISM's clinical + treatment-phase core reached about 0.70, and neither the radiomics branch, the frozen deep encoder nor a consistent re-segmentation added measurable discrimination. Third, the core model's signal is largely phase of care: whether and when RT started and ended. Within the post-RT surveillance phase no model, with or without imaging, discriminated well. Fourth, the audit layer met its coverage target and made each prediction traceable, but it did not improve discrimination beyond case mix, and its main value was to expose problems that the AUC hides.

What PRISM contributes. At AUC about 0.70 we did not find a triage benefit. Abstaining on uncertain scans and referring early post-RT scans raised the AUC of the remaining scans only by changing their phase mix, and the flags together covered most scans. The conformal sets met their marginal target, but they did so mainly by assigning "no progression" to pre-RT scans, missing every pre-RT progression, and coverage broke under the shift to LUMIERE. What PRISM does provide is auditability, phase awareness and an honest statement of uncertainty. Each record shows what the model knew at the scan date, where the scan sits in the treatment course, an interval and an exact explanation, and the report generator restates only what the record contains. Running the audit found a missingness shortcut, a look-ahead in the use of molecular results before diagnosis, and the cause of the external miscalibration, none of which the cross-validated AUC revealed. We think this is the honest use of such a system at this level of performance: to make a modest model's reasoning and failure modes visible, not to claim better decisions. The report generator carries this into the clinic's language. Grounded in the

record, it restated the record faithfully in all but 13 of 249 reports, all failing on one missing citation of a correct number, with no omitted flag and no forbidden content. The v1 prompt, given the same facts, issued RANO verdicts and management advice and dropped referral flags. Grounding the language model in an audit record, and checking it against that record, is what made the report auditable.

Why our initial evaluation misled. The single split did more than add noise. It changed the ranking question. When 60% of the case–control pairs compare a progression scan with a fresh post-operative scan before RT, any model that recognises a pre-RT scan scores well. The core model does this through treatment status. PRISM's initial radiomics model does it through T1c texture. A no-training "post-RT" rule scored 0.740 on that test set. This mechanism is not specific to radiomics. It also lifted a no-imaging model to 0.871, and, while future treatment dates still leaked into the features, a boosting model to 0.901. That model averaged about 0.70 under repeated CV. This agrees with the instability across random splits shown by An et al. (2021) and adds a specific mechanism: when cohorts mix phases of care with very different event rates, the composition of a small test set is a major source of variance, explaining about two thirds of this split's excess. A split search that targets the number of test scans and positives does not control it. We therefore suggest four reporting rules for this task: repeated grouped CV; the distribution over splits when a single split is reported; results within phases of care; and a clinical comparator that knows the treatment course.

Phase of care as a confounder. Event rates after a surveillance MRI depend strongly on where the scan sits in the course of treatment. A scan before RT is rarely followed by documented progression within 120 days. A scan in the early post-RT window often is, partly because of pseudoprogression and the follow-up imaging it triggers (Brandes et al., 2008; Taal et al., 2008; Wen et al., 2023). A four-level phase block recovered most of the core model's gain over clinical variables, treatment features alone reproduced the whole core model (0.701), and SHAP gave treatment status the largest share of credit. Two clinical comparators in the literature also use time since the end of RT and find it influential (Gomaa et al., 2025; Leone et al., 2023), though in a pseudoprogression task with an already suspicious lesion. Imaging models that pool pre-RT and surveillance scans can therefore report high AUCs for phase recognition rather than for progression biology. This is why PRISM makes phase of care an explicit part of every record and flags the early post-RT window. Our pre-specified test of imaging within the post-RT phase was negative (+0.061, −0.038 to +0.156), and part of its point estimate was an artefact of adding columns to a weak base.

What imaging did not add, and why. We found no pipeline defect: volumes matched the dataset's own table, labels were aligned and masks looked correct. Yet even at the scan where progression was documented, compartment volumes separated it from verified negatives with AUC only 0.68, and ET volume was no larger than at the previous scan in half of the pairs. Three features of the data explain this. The label is a documentation event: about one in eight

grade 4 events was a clinical-only progression, and the 120-day label mostly means that the next scan was the progression scan. Volume series were noisy, with changes over days as large as over months; a consistent segmenter reduced but did not remove this. And the cohort is small: by the Riley criteria even the 31-column core model would need about ten times more scans, and radiomics models far more (Riley et al., 2020). Cleaning the label and re-segmenting moved imaging in the expected direction by about +0.03, but neither met its pre-specified rule. A real imaging effect of that size cannot be excluded; our data cannot detect it, and PRISM keeps its imaging branches as optional modules so that the question can be re-tested on larger cohorts. Diffusion did not change the picture in CFB-GBM, although that test was small (19 events) and used precomputed features. The task literature where imaging performs well uses diffusion and perfusion for pseudoprogression classification (Kim et al., 2019; Taylor et al., 2022), a different question from forward risk at every surveillance scan. Imaging also took about half of the SHAP credit in models that included it, without any gain in discrimination: attribution shares among correlated features describe how a model divides its output, not whether a feature group is needed, so claims of imaging value need a paired difference with an interval.

External validity. Locked models lost about 0.05–0.06 AUC on LUMIERE, in line with the drops reported for externally validated imaging models (Yu et al., 2022). PRISM's initial radiomics model fell from 0.80 to 0.61. LUMIERE lacks treatment dates, so the treatment-aware core model could not be transported, and the shared model's under-prediction came mostly from a missingness indicator: IDH status is often unknown in LUMIERE, and in MU unknown IDH occurred only in scans without progression. The time to the next scan alone predicted the LUMIERE label better than any model, so external AUCs on this kind of label are partly schedule-dependent. In MU, by contrast, the core model kept its ranking within strata of the scan interval.

Comparison with other work. Our grade 4 core model (0.698) is close to the clinical comparators reported for adjacent tasks (0.69 in Leone et al., 2023; 0.727 externally in Gomaa et al., 2025) and to the nested-CV result for a volume-defined label on the same dataset (0.728; Rai, 2026 [CHECK]). It is below the single-test-set 0.80 reported on MU-Glioma-Post (Christodoulou et al., 2026), whose label (progression status of each scan) differs from our forward 120-day label; we did not re-evaluate that model. Higher AUCs for forward recurrence prediction elsewhere (for example Chen et al., 2026; Tak et al., 2025) use different cohorts, horizons and units, and some fall externally (0.887 to 0.762 in Chen et al., 2026).

Limitations

The development cohort is a single centre with 126 grade 4 patients and 93 events at 120 days, far below recommended sample sizes; small imaging effects cannot be excluded. The outcome is documented progression from routine records, not a blinded per-scan RANO read, and its timing depends on scheduling. MU images lack acquisition metadata, so scanner and field-strength

effects could not be audited, and race could not be studied. The external cohort is small (61 patients), uses a different segmenter and an expert RANO label, and lacks treatment dates, so PRISM's core model was not externally tested. Several analyses were added after earlier results were known (for example the clean-label and re-segmentation tests followed the imaging null), although each had its rule fixed before its own run; the phase-ladder plan was written into the code but not separately committed. Radiomics discretisation used a fixed bin count rather than the IBSI-recommended fixed bin width (Zwanenburg et al., 2020), and the ICC stability filter covered only T1c and FLAIR. Deep features were frozen; fine-tuning and diffusion-based deep features were not run. Feature pipelines in two secondary analyses (the feature-group search and one label-formulation set) still contain unclipped legacy radiomics. The LUMIERE score of PRISM's initial radiomics model predates the final external label definition. The audit found a small molecular look-ahead (8 grade 4 scans) that also affects the reported core model; its size was not quantified. The faithfulness checker is lexical: it verifies numbers, cited keys, flags, tier words and forbidden patterns, but not meaning (a correct number attached to the wrong concept), and its forbidden patterns produce false positives on negations; meaning was checked only by reading 15 reports. The report-generator primary was missed by one report; the obvious fix would be tuned on the evaluation set and needs fresh records to be tested. Reports came from one CV repeat of one cohort. Faithfulness to the audit record is not clinical usefulness: no clinician assessed the reports, and there was no reader study, so we cannot say whether the audit records or reports change clinical decisions.

Conclusion

PRISM's initial AUC of 0.80 reflected one favourable split and, for one of its comparators, a treatment-date leak; across splits its radiomics pipeline averaged 0.72, and its clinical + treatment-phase core reached about 0.70 under repeated cross-validation. MRI radiomics, frozen deep features, re-segmentation, a cleaner label and diffusion features added no measurable discrimination. Phase of care drives discrimination on this task and should be reported as a stratum and controlled as a confounder. PRISM is therefore deployed as its clinical + treatment-phase core, with imaging branches as optional modules. Its audit records and grounded reports make a modest model's knowledge and failure modes explicit, but on these data they did not make triage better than case mix.

Declarations

Ethics. [PENDING AUTHOR DECISION: secondary analysis of public de-identified datasets (MU-Glioma-Post, LUMIERE, CFB-GBM); state the institutional determination or waiver and cite the source datasets' approvals.]

Funding. [PENDING AUTHOR DECISION].

Competing interests. [PENDING AUTHOR DECISION].

Data availability. [PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83), LUMIERE (figshare; Suter et al., 2022) and CFB-GBM (TCIA, doi:10.7937/v9pn-2f72) are public; derived tables, fold assignments and locked model files — state what is released.]

Code availability. [PENDING: public repository link and release tag.]

Protocol and registration. No registered protocol. Analysis plans and dated plan snapshots are in the repository (Supplementary S1).

Patient and public involvement. None.

PRISM versions. PRISM v1 and its initial single-split evaluation were described in an unpublished internal report [PENDING AUTHOR DECISION: by whom; whether to cite it]; this manuscript supersedes it. "Our initial evaluation split" and "PRISM's initial radiomics model" refer to the split and the radiomics model of that report.

Tables

Table 1. Cohort characteristics. Index scans are post-treatment MRIs before first documented progression. Values are n (%) or median [IQR]. Patient rows use each patient's first index scan. Full table with definitions: Supplementary Table S2.

	MU all gliomas	MU grade 4	LUMIERE
Patients / index scans	155 / 308	126 / 249	61 / 132
Index scans per patient, median (range)	2 (1–6)	2 (1–6)	2 (1–10)
Age, years	61 [47–69]	62 [53–70]	62 [53–70]
Male sex	80 (52%)	66 (52%)	32 (52%)
IDH wild type / mutant / unknown	120 (77%) / 22 (14%) / 13 (8%)	112 (89%) / 8 (6%) / 6 (5%)	38 (62%) / 1 (2%) / 22 (36%)
MGMT methylated / unmethylated / unknown	53 (34%) / 73 (47%) / 29 (19%)	45 (36%) / 62 (49%) / 19 (15%)	26 (43%) / 28 (46%) / 7 (11%)
WHO grade 4	126 (81%)	126 (100%)	61 (100%)
Radiotherapy documented	102 (66%)	89 (71%)	not recorded
Initial chemotherapy documented	96 (62%)	88 (70%)	not recorded
Days from origin to scan[a]	97 [46–186]	96 [54–182]	119 [82–226]
RT completed by scan day (scans)	185 (60%)	167 (67%)	not recorded
Early post-RT window[b] (scans)	104 (34%); unknown 94	98 (39%); unknown 65	27 (20%)
Gap to next MRI, days	77 [50–103]	74 [49–98]	91 [77–117]
Death without documented progression (scans)	22 (7%)	20 (8%)	7 (5%)
120 d: progression / no progression / censored	100 / 156 / 52	93 / 117 / 39	47 / 76 / 9
90 d	65 / 195 / 48	62 / 152 / 35	24 / 101 / 7
180 d	144 / 105 / 59	131 / 75 / 43	62 / 61 / 9

[a] MU: days since diagnosis; LUMIERE: days since first surgery.
[b] MU: 0–84 days after the documented RT end date; LUMIERE: rater's "< 3 months after RT" flag.

Table 2. The same models on a single split, across 500 splits and under repeated cross-validation. IPCW AUC(120 d) with 95% patient-cluster bootstrap CI unless stated. Single split = our initial evaluation split (all gliomas; 29 test patients, 82 scans). 500 splits = PRISM v1's split design (naive label; for PRISM's initial radiomics model, its pipeline with features re-selected per split). Core model = PRISM's deployed clinical + timing + as-of-scan treatment model (the baseline of every imaging comparison). 5 × 5 CV = repeated patient-grouped CV on the label-fixed cohort. History rows document the as-of-scan treatment correction on the earlier 30-patient / 84-scan version of the split; "pre-fix (leaky)" means that future RT dates and RT status entered the treatment and timing features.

Model	Single split: naive AUC	Single split: IPCW AUC [95% CI]	Δ vs core ridge [95% CI]	500 splits: naive mean ± SD (2.5–97.5%)	5 × 5 CV, all gliomas	5 × 5 CV, grade 4
PRISM initial radiomics model	0.802	0.817 [0.690, 0.920]	−0.034 [−0.186, +0.109]	0.720 ± 0.055 (0.610–0.819)	n/a (fixed model)	n/a (fixed model)
Core model (ridge)	0.871	0.851 [0.735, 0.947]	–	0.801 ± 0.052 (0.694–0.895)	0.720 [0.638, 0.792]	0.698 [0.615, 0.774]
Core model (GBM)	0.844	0.824 [0.690, 0.916]	−0.027 [−0.127, +0.062]	not run	0.718 [0.642, 0.792]	0.687 [0.601, 0.768]
Core + legacy radiomics	0.862	0.845 [0.746, 0.936]	−0.006 [−0.081, +0.090]	0.799 ± 0.053 (0.674–0.892)	0.743 [0.678, 0.803]	0.703 [0.627, 0.774]
Clinical only (ridge)	0.791	0.782 [0.638, 0.897]	−0.069 [−0.215, +0.058]	not run	0.619 [0.540, 0.696]	0.585 [0.496, 0.666]
History rows (30 patients / 84 scans; earlier cohort and inputs)						
PRISM initial radiomics model	0.802	0.811 [0.694, 0.919]	not computed	–	–	–
Core ridge, pre-fix (leaky)	0.870	0.844 (no CI)	not computed	not run	0.719 [0.648, 0.788][c]	0.694 [0.613, 0.775][c]
Core ridge, after as-of-scan fix	0.880	0.860 [0.730, 0.943]	not computed	not run	0.749 [0.675, 0.820][d]	0.729 [0.648, 0.811][d]
Core GBM, pre-fix (leaky)	0.901	0.881 (no CI)	not computed	not run	0.743 [0.666, 0.816][c]	0.723 [0.633, 0.806][c]
Core GBM, after as-of-scan fix	0.869	0.855 (no CI)	not computed	not run	0.740 [0.663, 0.815][d]	0.731 [0.641, 0.813][d]

[c] Pre-fix inputs, pre-label-fix cohort, earlier repeat-averaged estimator.
[d] Intermediate inputs after the as-of-scan fix, before the label fixes; repeat-averaged estimator. History-row CV values are not comparable with the final rows.

Table 3. Do PRISM's imaging branches add to its core model? Paired ΔAUC with 95% patient-cluster bootstrap CI. Reference: PRISM's core model (clinical + timing + treatment, ridge) unless stated in the section heading. Grade 4 (G4) is primary. * = CI excludes 0 (uncorrected; 3 of the 32 intervals in Figure 3B; the CFB-GBM row is added here).

Analysis	G4 ΔAUC [95% CI]	All gliomas ΔAUC [95% CI]
Radiomics branch vs core model		
Nested over 70 configurations (pre-specified primary)	−0.032 [−0.084, +0.017]	+0.028 [−0.017, +0.071]
Legacy top-10, concatenated (single configuration)	+0.005 [−0.059, +0.067]	+0.024 [−0.026, +0.075]
Legacy PCA-10, stacked (best post hoc, optimistic)	+0.023 [−0.013, +0.057]	+0.035 [+0.002, +0.068]*
Frozen deep encoder branch vs core model		
Nested over 32 (BraTS21 SwinUNETR + BrainSegFounder)	−0.004 [−0.069, +0.063]	+0.011 [−0.032, +0.056]
BraTS21 SwinUNETR, lesion PCA-10 stacked	+0.016 [−0.029, +0.059]	+0.017 [−0.014, +0.050]
BrainSegFounder, lesion PCA-10 stacked	+0.014 [−0.029, +0.057]	+0.024 [−0.013, +0.064]
Random-weight encoder, nested (control)	−0.049 [−0.085, −0.014]*	−0.009 [−0.032, +0.009]
Imaging-defined label (clinical-only progression censored)		
Core + volumes/change vs core (pre-specified rule)	+0.027 [−0.033, +0.082]	+0.008 [−0.036, +0.049]
Post-RT scans only		
Clinical + phase + imaging vs clinical + phase (pre-specified primary)	+0.061 [−0.038, +0.156]	+0.033 [−0.054, +0.128]
Core + imaging measurements vs core	+0.017 [−0.069, +0.093]	+0.005 [−0.075, +0.082]
Core + radiomics (nested) vs core	+0.005 [−0.070, +0.075]	+0.024 [−0.030, +0.076]
Re-segmentation (reference: same model on MU masks)		
Clinical + phase + imaging, BraTS21 SwinUNETR masks	+0.030 [−0.008, +0.069]	+0.029 [−0.010, +0.069]
Clinical + phase + imaging, BrainSegFounder masks	+0.008 [−0.023, +0.040]	+0.003 [−0.025, +0.032]
LUMIERE, locked (reference: clinical + timing)		
+ volumes	−0.001 [−0.073, +0.057]	−0.006 [−0.075, +0.055]
+ deep features	+0.057 [−0.132, +0.235]	+0.064 [−0.011, +0.144]
+ volumes + deep vs + volumes	+0.089 [−0.026, +0.215]	+0.050 [+0.002, +0.109]*
CFB-GBM (77 patients, 19 events; reference: conventional radiomics)		
+ ADC radiomics (pre-specified primary)	+0.007 [−0.019, +0.035] (permutation p = 0.151)	–

Figure legends

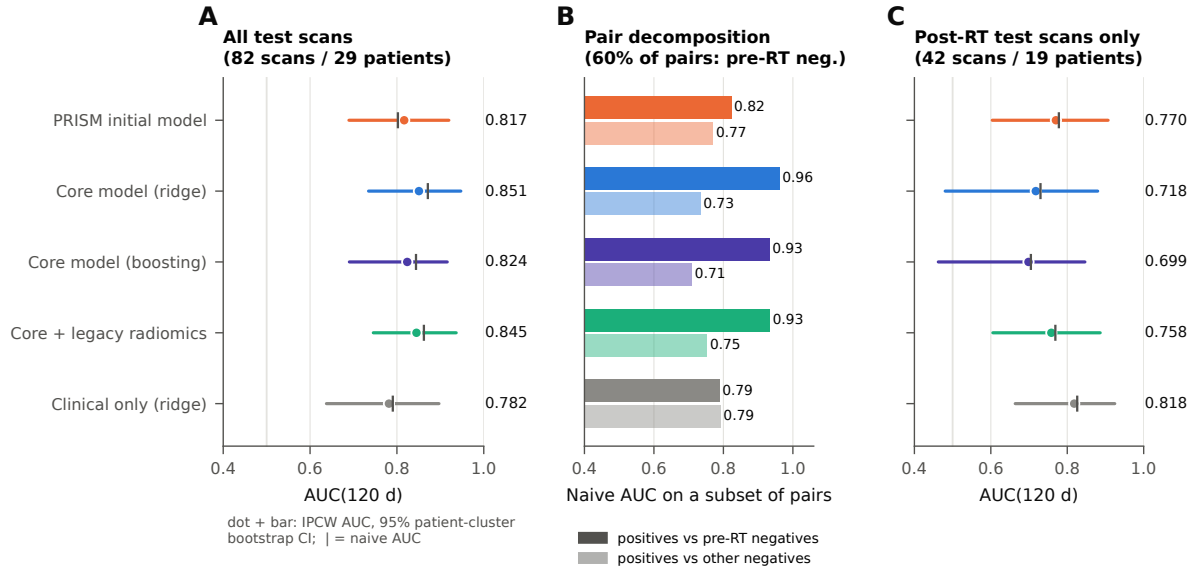


Figure 1. PRISM's initial single-split result. Discrimination on our initial evaluation split (all gliomas; 126 training / 29 test patients, 82 test scans, 22 naive positives). Every model except PRISM's initial radiomics model was fitted once on the training patients; the initial radiomics model enters with its shipped test-set probabilities. (A) IPCW AUC with 95% patient-cluster bootstrap CI (1,000 draws; dot and bar) and naive AUC (tick). (B) Naive AUC decomposed by the type of negative scan: 36 of the 60 negative test scans were taken before RT started, and none of them progressed, so 60% of positive–negative pairs compare a progression scan with a pre-RT scan; solid bars, positives against pre-RT negatives; light bars, positives against other negatives. (C) The same models on the 42 post-RT test scans (19 patients). Intervals are wide and overlap throughout.

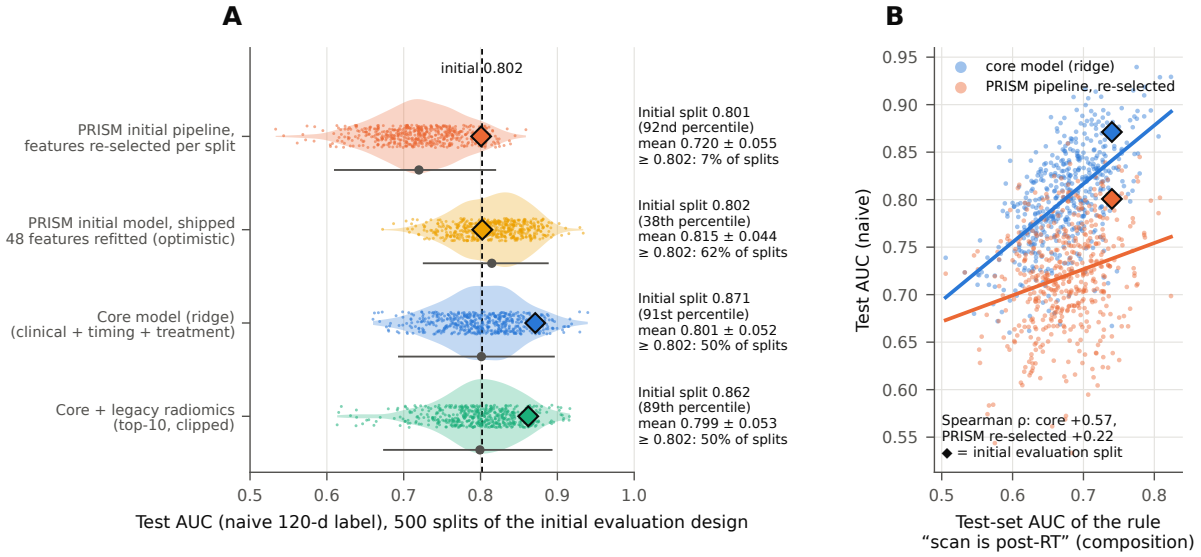


Figure 2. Where our initial evaluation split sits among 500 splits of the same design. PRISM v1's split search was run for seeds 0–499; seed 62 reproduces our initial evaluation split. (A) Test AUC (naive label) per split for PRISM's initial radiomics pipeline with features re-selected per split (honest estimate), the shipped features refitted (optimistic upper bound), the core ridge model and the core model + legacy radiomics. Violin and points, 500 splits; grey dot and line, mean and 2.5–97.5% range; diamond, each model's own value on our initial evaluation split with its percentile rank; dashed line, the reported 0.802. (B) Per-split AUC against the AUC of the no-training rule "scan is post-RT" on that test set (higher = more pre-RT negatives and fewer pre-RT events); lines, least-squares fits; Spearman ρ +0.57 (core model) and +0.22 (initial radiomics pipeline).

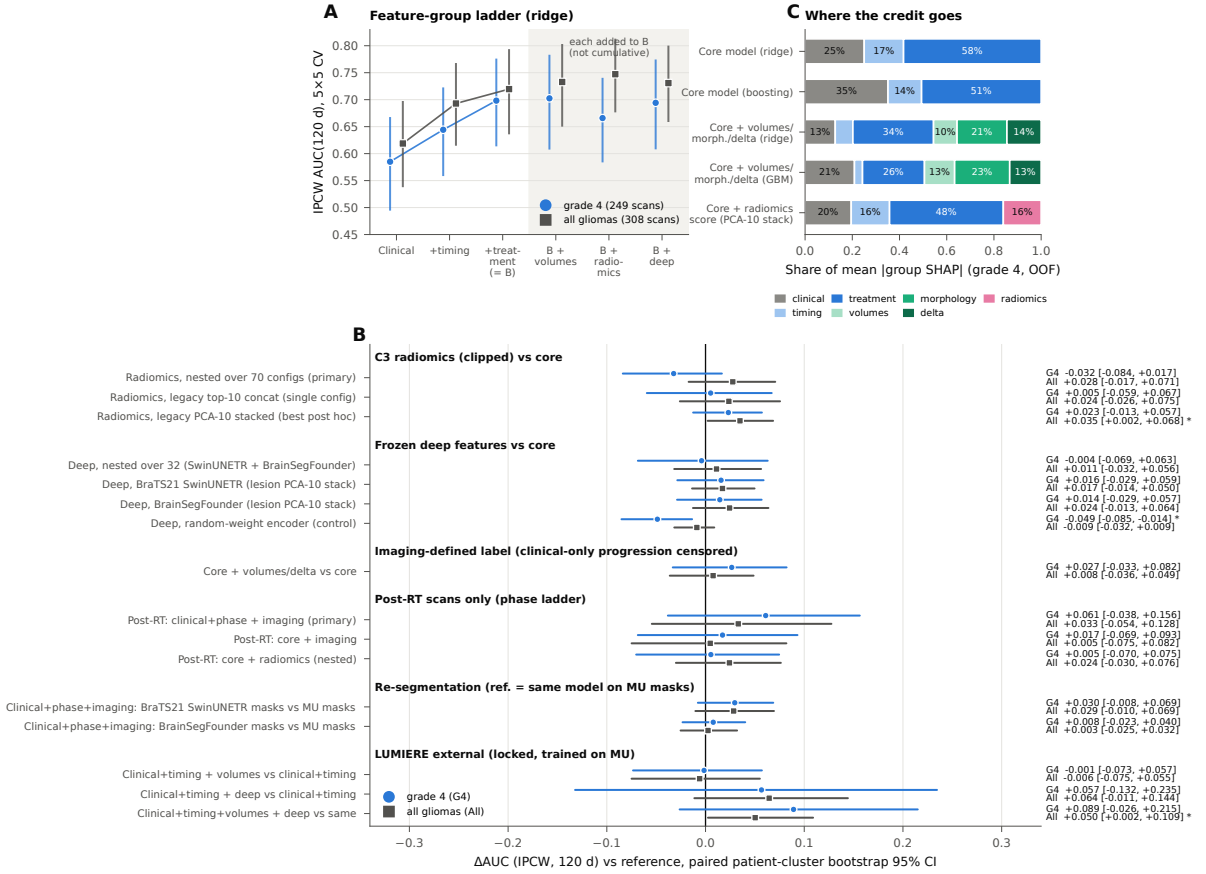


Figure 3. Component ablation: clinical, treatment-phase and imaging branches under repeated cross-validation. (A) Feature ladder (ridge; 5 × 5 repeated patient-grouped CV; IPCW AUC(120 d) with 95% patient-cluster bootstrap CI): clinical → + timing → + as-of-scan treatment (= PRISM's core model, B); the last three points each add one imaging branch to B (volumes; radiomics, nested over 70 configurations; frozen deep features, nested over 32). (B) Paired ΔAUC with 95% CI against the reference named in each section (Table 3). Asterisks mark CIs excluding 0 (3 of 32, uncorrected; about 1.6 expected by chance). (C) Share of mean absolute group SHAP in grade 4 out-of-fold explanations: imaging blocks take 45–51% of the credit in the full models without improving discrimination.

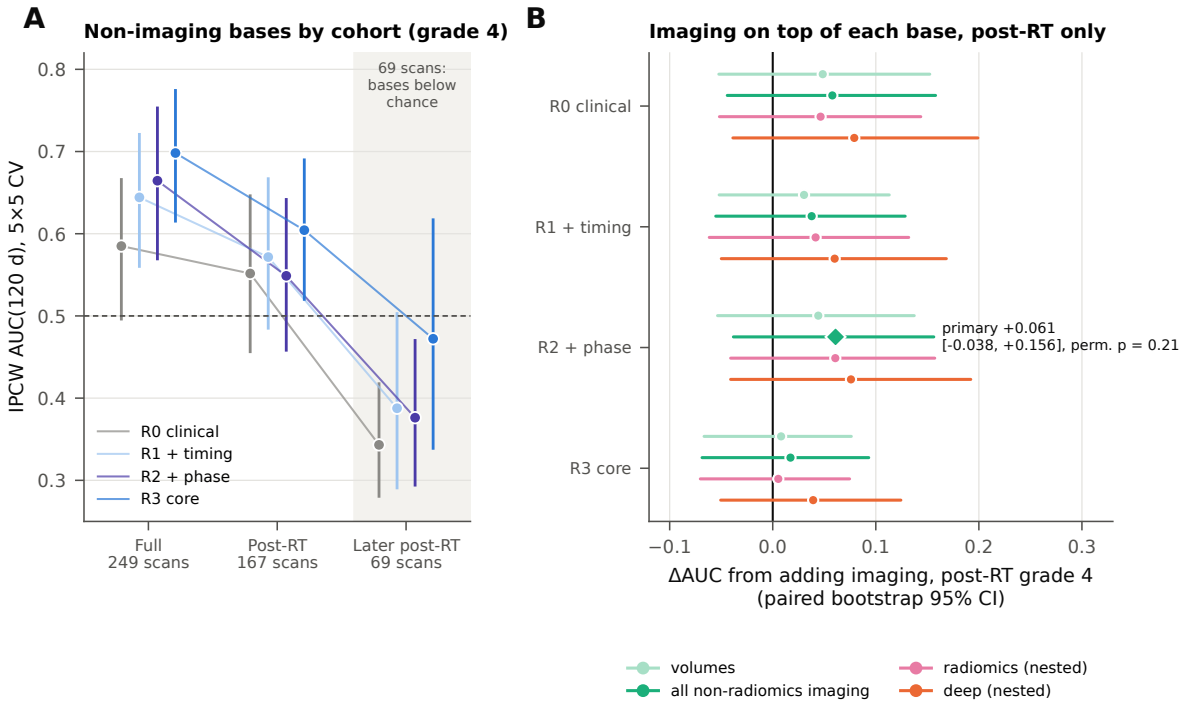
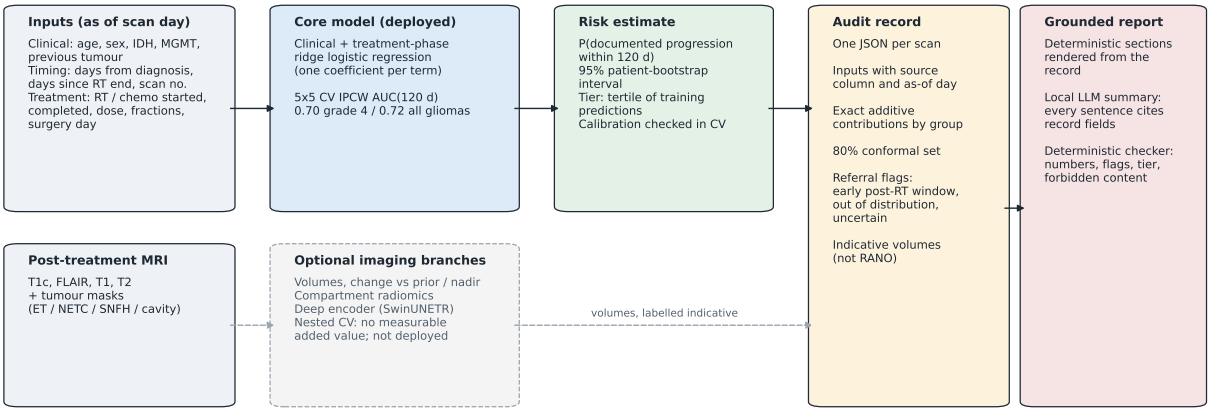


Figure 4. Phase of care. (A) Non-imaging bases (ridge, grade 4) in the full cohort (249 scans), the post-RT cohort (167 scans, 84 patients, 83 events) and the later post-RT cohort (> 12 weeks after RT end; 69 scans), each trained and evaluated within the cohort. R0 clinical; R1 + timing; R2 + four-level phase of care; R3 core model. In the later post-RT cohort the bases fall below 0.5, the expected anti-learning of a near-null model in a small sample, so contrasts there are not interpretable. (B) ΔAUC from adding each imaging block (I1 volumes; I2 all non-radiomics measurements; I3 radiomics; I4 deep) to each base within grade 4 post-RT scans. Diamond, pre-specified primary contrast (R2 + I2 vs R2): +0.061 (−0.038 to +0.156), permutation p = 0.21; the permutation null itself averages +0.03.

A



B

PRISM surveillance report

PatientID_0003_Timepoint_1 | scan day 90 from diagnosis | early post-RT (0-84 days after the end of radiotherapy)

Risk of documented progression within 120 days

0.50 (95% interval 0.26-0.69)

Tier: middle third of development-cohort predictions (cut points 0.30 / 0.53). 80% conformal set: both outcomes (progression and no progression).

Referral flags

[!] early post-RT window: scan 25 days after the end of RT, inside the 84-day early post-RT window where treatment effects can mimic progression

[!] out of distribution: MGMT status unknown

[!] uncertain: the 80% conformal prediction set contains both outcomes

Action: flagged for clinician review

Contributions (exact additive log-odds; positive = towards higher risk)

Intercept -0.20 clinical +0.36 timing -0.41 treatment +0.26 = logit +0.01. Imaging: not part of this model.

-0.17 scan number = 1 (timing)
-0.14 within 84 days after RT end: yes (timing)
+0.12 male sex: no (clinical)
+0.11 day of first surgery = -5 (treatment)
+0.11 previous brain tumour: no (treatment)

Inputs as of the scan day (source: MU clinical workbook)

Age 57, female; IDH wildtype; MGMT unknown; RT started yes, completed yes; days since RT end 25; chemotherapy started yes. All 23 inputs known by the scan day: yes (each with source column and as-of day in the record).

Volumetric change (indicative, not RANO)

Enhancing 33.8 cc; FLAIR hyperintensity 44.3 cc.

First scan: no prior or nadir.

Summary (local LLM; every sentence cites record fields; checked deterministically)

This scan was performed 90 days from diagnosis, 25 days after the end of radiotherapy, during the early post-RT phase of care.

The model estimates a 50% risk of documented progression within 120 days, with a 95% interval ranging from 26% to 69%, placing the patient in the middle third of risk.

The 80% conformal prediction set includes both progression and no progression outcomes, indicating uncertainty in the specific trajectory.

The scan is flagged for clinician review because it falls within the early post-RT window where treatment effects can mimic progression, MGMT status is unknown, and the prediction set is uncertain.

Indicative imaging volumes, which are not a RANO assessment, show an enhancing volume of 33.8 cc and FLAIR hyperintensity of 44.3 cc.

This tool is intended for decision support alongside clinical and radiological assessment and does not provide a diagnosis.

Faithfulness check: 6/6 claims supported; active flags mentioned: all; forbidden content: none.

Limitations

Model discrimination in development cross-validation AUC 0.70 (single institution, MU-Glioma-Post); the risk is the probability of documented progression within 120 days, not a diagnosis of progression or pseudoprogression; volumes are indicative, not a RANO assessment.

C

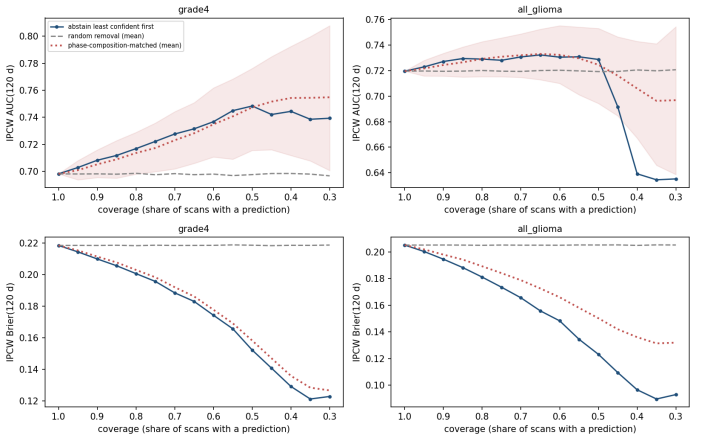


Figure 5. PRISM: architecture, grounded report and selective prediction. (A) Architecture and data flow: inputs read as of the scan day feed the deployed core model (clinical + treatment-phase ridge); the imaging branches (volumes, radiomics, deep encoder; dashed) are optional and not deployed, and volumes enter the record only as indicative values; the risk, with its interval and tier, goes into the per-scan audit record (contributions, 80% conformal set, referral flags), from which the grounded report is written. (B) Grounded report for the first early post-RT evaluation scan (chosen by a pre-specified rule): deterministic sections rendered from the record and a local language-model summary in which every sentence cites record fields; all three active flags are stated and the checker found every claim supported. (C) Risk–coverage curves for the core model (5×5 CV, pooled out-of-fold predictions, mean over repeats), grade 4 (left) and all gliomas (right). Top, IPCW AUC(120 d); bottom, IPCW Brier score, of the scans retained when the least confident scans (lowest $\max(p, 1 - p)$) are withheld first (solid), against random removal (dashed) and phase-composition-matched random removal (dotted; shaded band, 2.5–97.5% of the composition-matched draws). The gain over random removal is almost entirely matched by the composition-matched reference (pre-specified P2 at about 70% coverage: +0.006, −0.023 to +0.036, grade 4).

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